

# Sinu Baby

📍 Chennai    📞 9744184493    ✉️ sinuarlowbaby.dev@gmail.com    💻 sinuarlowbaby    🌐 github.com/sinuarlowbaby

## SUMMARY

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MCA candidate specializing in Generative AI, retrieval engineering, and AI backend systems. Experienced in building RAG pipelines, agentic AI workflows, and LLM-powered applications using FastAPI, LangChain, LangGraph, Qdrant, Redis, and PostgreSQL. Focused on hybrid retrieval, semantic search optimization, async API architecture, and Dockerized backend infrastructure.

## TECHNICAL SKILLS

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**Languages:** Python (Advanced), SQL, JavaScript, HTML5, CSS3

**AI / GenAI:** LLMs, RAG, Agentic AI, LangChain, LangGraph, Prompt Engineering, Semantic Search, Hybrid Retrieval, Cross-Encoder Reranking, Vector Embeddings, Semantic Caching, AI Workflow Orchestration

**Backend & Frameworks:** FastAPI, Django, REST APIs, Async Python, SQLAlchemy, Pydantic, JWT, OAuth

**Databases:** PostgreSQL, MySQL, SQLite, ChromaDB, Qdrant, Redis/Valkey

**DevOps & Tools:** Docker, Docker Compose, Git, GitHub, Postman, VS Code, Kubernetes

## PROJECTS

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### **RAG-PDF-Chatbot** | *FastAPI, Qdrant, Redis, LangChain, Groq, Docker*

- Engineered a Retrieval-Augmented Generation (RAG) system for large-scale PDF querying using FastAPI, Qdrant, Redis semantic caching, and LangChain orchestration.
- Designed a multi-stage RAG pipeline using semantic chunking, hybrid retrieval (MMR + BM25), metadata filtering, and cross-encoder reranking to improve retrieval relevance and grounding consistency.
- Reduced redundant LLM response latency by ~35% and minimized API inference costs by implementing Redis semantic caching and token-aware context assembly.
- Implemented SSE streaming architecture, async inference pipelines, and Dockerized deployment workflows for real-time AI response generation.
- Integrated OpenAI embeddings and vector indexing strategies in Qdrant for efficient semantic retrieval across multi-document datasets.

### **Agentic AI Chatbot** | *LangGraph, FastAPI, GPT-model, PostgreSQL, Tavily*

- Built a multi-step autonomous AI agent using LangGraph with Plan → Execute → Evaluate orchestration workflows.
- Implemented tool-based reasoning workflows for web search, contextual analysis, and summarization using LangChain, GPT-4o, and Groq LLaMA models.
- Developed FastAPI APIs with Pydantic structured outputs, PostgreSQL conversation tracking, and conditional graph routing.

### **CareerLens AI** | *FastAPI, LangChain, ChromaDB, Groq LLaMA 3.3, PostgreSQL, Docker*

- Built an AI career coaching platform with ATS scoring, resume-job gap analysis, and mock interview system using FastAPI, LangChain, ChromaDB, and Groq LLaMA 3.3.
- Implemented RAG pipelines with SemanticChunker, vector search, structured LLM outputs, JWT + Google OAuth, PostgreSQL integration, and Dockerized scalable backend architecture.
- Designed semantic retrieval workflows and structured evaluation pipelines to improve contextual resume-job matching and interview response analysis.

## EDUCATION

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### **SRM Institute of Science and Technology**

*Master of Computer Applications (MCA) | Specialization in Generative AI*

Kattankulathur

2025 – 2027

### **CM College of Arts and Science, Calicut University**

*Bachelor of Computer Applications (BCA)*

Nadavayal

2022 – 2025

## ADDITIONAL INFORMATION

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**Languages:** English, Malayalam, Hindi, Tamil

**Internship:** AICTE Internship in Generative AI

**Interests:** Active Participant in Intercollegiate Hackathons and Development Challenges